<u>Final Project Assignment</u>

Course: Applied Python Programming for Data Science
Maximum Marks: 40 (Note: Marks will be awarded proportionately for each component. If any component is not applicable, provide a clear justification.)

Objective: Design and implement an end-to-end data science project using Python on a time series dataset from an open-source repository (e.g., domain-specific datasets or Kaggle).

1. Data Acquisition and Preprocessing

- Selection of a raw, real-world time series dataset
- Handling missing values, noise, and inconsistencies
- Data transformation and normalization (if required)

2. Exploratory Data Analysis (EDA)

- Visualization of temporal patterns
- Identification of trends, seasonality, and anomalies
- Interpretation of variable relationships

3. Feature Engineering

- Creation of time-based and domain-specific features
- Lag variables, rolling statistics, or transformations
- Justification of feature choices

4. Model Development

Implement and compare the following models:

- Statistical Models: ARIMA / SARIMAX
- Machine Learning Model: XGBoost
- Deep Learning Model: LSTM

5. Model Evaluation and Comparison

- Use appropriate error metrics (e.g., MAE, RMSE)
- Compare model performance across methods
- Analyze strengths and limitations of each approach

7. Reporting and Reproducibility

- Well-documented Python notebook (clear code + explanations)
- Visualization of results
- Clear conclusions and possible improvements